APOHARA · CONSILIUM

The Agent Governance OS for Autonomous AI

9-vendor adversarial ensemble · INV-15 Z3 formal proof · RFC 3161 TSA timestamps
DORA Art 9/12/13 + EU AI Act Art 14 Aug-2026 ready

STABLE v1.0 | Apache-2.0 | DOI 10.5281/zenodo.20277875 | Milan AI Week 2026

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Italian banks need DORA + EU AI Act compliance evidence by Aug 2.

THE REGULATORY MANDATE

- DORA mandatory since 2025-01-17 — 22K+ EU financial entities; UniCredit + Intesa Sanpaolo as G-SIBs forced to comply. ICT risk management + audit trail requirements.
- EU AI Act Art 14 (human-oversight obligation) — fully enforceable 2026-08-02 (~75 days · penalties up to €35M or 7% global revenue).
- Italian regulators: Banca d'Italia, CONSOB, IVASS, ACN — all issuing AI governance guidance aligned to DORA + EU AI Act.
- No audit trail = no compliance evidence. Autonomous AI agent workflows require tamper-evident logs under both frameworks.

THE TECHNICAL GAP

- OWASP LLM Top-10 2026 elevated **Tool Poisoning to LLM02** on 2026-04-14 — AI agents need independent veto layers.
- Self-review is same-model — same training data, same blind spots, same hallucinations. Cross-vendor adversarial ensemble closes this.
- CSA Agentic Profile (March 2026 DRAFT) extends NIST AI RMF for autonomous agents. NIST "Agent Interoperability Profile" — Q4 2026 planned.
- RFC 3161 TSA timestamps + HMAC-SHA256 chain → tamper-evident audit trail aligned to DORA ICT Art 9/12/13 requirements.

9-vendor adversarial ensemble + court-grade audit trail.

A — OSS ENTRY

Apache-2.0 · free forever

- 9-vendor adversarial ensemble (Claude · GPT · DeepSeek · Kimi · GLM · Qwen · Nemotron · MiniMax · Mistral)
- 78-rule DJL deterministic judge · TPR/TNR 1.000 · p99 0.114 ms
- INV-15 Z3 SMT formal proof · UNSAT in 10.08 ms · 0/1210 empirical sweep

B — GOVERNANCE OS CORE

"Splunk for AI agents"

- 4-stage SOAR pipeline · DETECT → JUDGE → ENFORCE → FORENSICS · p99 10.6 ms
- DORA + EU AI Act compliance dashboards · 6 frameworks · 14 mapped controls
- HMAC-SHA256 verdict chain + STIX 2.1 export for SIEM integration

D — CAICEP MODULE

Court-grade audit trail · roadmap

- RFC 3161 TSA timestamps (freetsa.org · 2026-05-19T12:21:50+00:00 live evidence) — tamper-evident, not yet court-admissible
- Roadmap: eIDAS QTSP partnership Q3 2026 → legally binding qualified timestamps under EU Regulation 910/2014
- DORA Art 9/12/13 aligned ICT audit log · Banca d'Italia / CONSOB / IVASS ready

9 frontier vendors · 4-stage SOAR · INV-15 Z3 proof · RFC 3161 timestamps

HOW THE MODELS COMPOSE

- Gemini 3.1 PRO — writes the review (BYOK or demo key, 5/IP/day).
- 9-vendor adversarial ensemble via OpenRouter + native APIs (Claude Opus 4.7, GPT-5.5, DeepSeek V4 Pro, Kimi K2.6, GLM 5.1, Qwen 3.6 Plus, Nemotron 3 Super 120B, MiniMax M2.7, Mistral Large 2411).
- HMAC-SHA256 chain + RFC 3161 TSA — every verdict timestamped (freetsa.org · live evidence: 2026-05-19T12:21:50+00:00). Tamper-evident; roadmap to eIDAS QTSP Q3 2026.
- Verdict combine — DJL ⊥ LLM ensemble in parallel via asyncio.gather, safe-merge BLOCK∨BLOCK / ALLOW∧ALLOW / else REVIEW. Equal veto.

THE INFRASTRUCTURE BENEATH

- Apohara ContextForge — KV-cache coordination layer for vLLM, hardware-validated on AMD Instinct MI300X (192 GB HBM3, ROCm 7.x). INV-15 Z3 UNSAT 10.08 ms ±0.5.
- Veea LobsterTrap DPI — pre-LLM regex interception (50% SQLi block, 9.8% FPR measured).
- Zero-LLM Deterministic Judge Layer — 78 regex rules (8 categories, bilingual EN+ES), **p99 0.114 ms**, TPR/TNR 1.000, prompt-injection-immune by construction.
- STIX 2.1 export at /v1/soar/incidents/{id}/stix — 7-SDO bundle for SIEM integration. DORA Art 9/12/13 ICT audit log alignment.

The only OSS combining all four of these.

① FORMAL PROOF

INV-15 (Judge-Critic-Responder gate) encoded as Z3 SMT problem; UNSAT on negation in 10.08 ms. Complements the empirical 0/1210 sweep from V2.0.1.

No competitor has a formal safety invariant. Paper v3.0 on Zenodo DOI 10.5281/zenodo.20277875.

② 9-VENDOR ENSEMBLE

Multi-vendor adversarial diversity: 9 active frontier vendors (Apache-2.0, self-hostable).

PLAYBOOK SOAR (Gemini-only), Pantheon (Gemini-only), Vela (Gemini-only), Trusyn (Gemini multi-variant) — all single-vendor.

Threshold ladder auto-scales {high:9, med:6, human_review:3}.

③ DUAL-LAYER VETO

DJL + LLM ensemble run in parallel with equal veto power: BLOCK ∨ BLOCK, ALLOW ∧ ALLOW, else REVIEW.

PLAYBOOK SOAR uses DJL as primary gate. We give the LLM ensemble paritetic authority — strictly safer.

④ MYTHOS-READY ARCHITECTURE (INDUSTRY-FIRST POSITIONAL GESTURE)

A reserved 13th attacker slot for Claude Mythos — Anthropic's most advanced model — wired through Apohara PROBANT's adversarial ensemble against itself. Currently INACTIVE until Claude for Open Source / Project Glasswing approval and credential provisioning. Honest boundary text in MYTHOS_READY.md; no access claim.

6 frameworks · 14 mapped controls
EU AI Act · NIST AI RMF · ISO 42001 · SOC 2 · GDPR · NIST 800-53

INDUSTRY TEMPLATES

- Finance — DORA Art 9/12/13 · PCI-DSS 4.0 · SOX · EU MiFID II · NIST SP 800-53 · UniCredit / Intesa Sanpaolo G-SIB scope
- Healthcare — HIPAA Privacy + Security Rules · GDPR Art. 9 · ISO 13485
- Government — EO 13526 · NIST SP 800-53 · FedRAMP · classified-marking handling
- Retail — PCI-DSS 3.2/3.4 · cardholder-data storage prohibition
- Manufacturing — IEC 62443 OT/ICS · NERC CIP for energy-adjacent
- Energy — NERC CIP-007 · grid-control directive interception

COMPLIANCE FRAMEWORKS (6 · 14 MAPPED CONTROLS)

- EU AI Act — Art 14 human-oversight. Fully applicable 2026-08-02 (~75 days · €35M / 7% revenue penalty).
- NIST AI RMF — GOVERN / MAP / MEASURE / MANAGE functions. CSA Agentic Profile extension (March 2026 DRAFT).
- ISO 42001 — AI management system. Audit-trail clause alignment via RFC 3161 + HMAC chain.
- SOC 2 — CC6.1 / CC6.6 / CC7.2 / CC7.3 (logical access + change management + incident response).
- GDPR — Art 22 automated decision-making · Art 30 records of processing · DPIA alignment.
- NIST SP 800-53 — AC / AU / SI / SC family controls for agentic AI systems.

Honesty discipline: every digit cites a log.

DJL P99 LATENCY

0.114 ms

76 regex rules · 1000 iterations · 124-prompt corpus. Bench: logs/djl_latency.json. 44× under 5 ms budget.

JBB BEHAVIORS BLOCK

93.75%

75 / 80 NeurIPS 2024 holdout · Wilson 95% CI [86.2%, 97.3%]. Day-5 10-frontier baseline log committed.

SOAR P99 LIFECYCLE

10.6 ms

DETECT→JUDGE→ENFORCE→FORENSICS · 1000 iter. logs/lifecycle_latency.json. 19× under 200 ms.

TESTS PASSING

603 / 0

Full aegis fusion-sprint regression in 33.61 s · 1 skip pre-existing · brand + honesty CI gates exit 0.

THE HONESTY CONTRACT — COMMITTED CI GATES

- ✓ check_honesty_fusion.sh — 4 rules: Mythos boundary · first-impl backing · test-count consistency · no "v2".
- ✓ check_brand_fusion.sh — bans the 10-hex PLAYBOOK SOAR teal palette.
- ✓ check_submission_lengths.sh — char-gate (4 submission variants < 2000 chars long-desc).
- ✓ scripts/check_honesty.sh — bans hardcoded latency / VRAM fabrication.

WHAT WE ARE NOT

- ✗ Not a generic LLM judge — IBM Granite Guardian 4 beats us on prompt-only classification (98.75% vs 93.75%, same JBB holdout). We measured it, we did not hide it.
- ✗ Not a content classifier — DJL is OWASP LLM 2026 + structured policy; we added a HARM category to cover the gap.
- ✗ Not Mythos-active — explicit boundary text in MYTHOS_READY.md: "Apohara has NOT been granted Mythos access."

Five layers of defense.
Two layers of veto.

```
          Hero / SDK / curl  (browser · LangChain · CrewAI)
                      │  POST prompt
                      ▼
┌────────────────────────────────────────────────────────────────┐
│ ① LobsterTrap DPI       Veea regex interception · pre-LLM     │
│ 50% SQLi block · 9.8% FPR · ~12 ms                           │
└────────────────────────────────────────────────────────────────┘
                      ▼
┌────────────────────────────────────────────────────────────────┐
│ ② Zero-LLM DJL          76 regex rules · 8 categories         │
│ TPR / TNR 1.000 · Wilson [0.9962, 1.0000] · p99 0.114 ms        │
└────────────────────────────────────────────────────────────────┘
                      ▼
┌────────────────────────────────────────────────────────────────┐
│ ③ 4-stage SOAR pipeline     DETECT → JUDGE → ENFORCE → FORENSICS │
│                                                               │
│  ┌── asyncio.gather (parallel, equal veto) ──────────────────┐   │
│  │  ④a DJL deterministic          ④b LLM ensemble (Gemini)  │  │
│  └──────────────────────────────────────────────────────────┘   │
│                          ▼                                      │
│ verdict_combine:  BLOCK ∨ BLOCK · ALLOW ∧ ALLOW · else REVIEW    │
│ p99 lifecycle 10.6 ms · 19× under 200 ms target                │
└────────────────────────────────────────────────────────────────┘
                      ▼
┌────────────────────────────────────────────────────────────────┐
│ ⑤ Apohara ContextForge   INV-15 KV-cache isolation            │
│ Z3 SMT formal proof · UNSAT in 10.08 ms · empirical 0 / 1210    │
│ 14 attackers, each in its own isolated KV-cache                 │
└────────────────────────────────────────────────────────────────┘
                      ▼
┌────────────────────────────────────────────────────────────────┐
│ ⑥ HMAC-SHA256 verdict chain  verify_chain() audit             │
│ STIX 2.1 export at /v1/soar/incidents/{id}/stix               │
└────────────────────────────────────────────────────────────────┘
```

Production, not slides. Click any URL.

WHAT A JUDGE CLICKS (60 SEC)

PATH	WHAT IT SHOWS
www.apohara.dev	Landing + Hero verify panel (BYOK or 5/IP/day demo)
/dashboard	Live verdict trend · SOAR metrics tiles · 11-route sidebar
/judge-layer	Dual-panel DJL + LLM ensemble · live BLOCK on bad prompts
/compliance	6 framework cards · 1-click Gemini compliance report
/simulator	3 demo agents × 9 misbehavior scenarios + Wilson CI health
/analytics	DJL block-rate sparkline · 6-framework incident counts
/review-queue	REVIEW-verdict triage view · BLOCK/ALLOW override actions

API SURFACE (8 SOAR ENDPOINTS + STIX EXPORT)

```
# DJL hits the worst prompt in 0.077 ms — no LLM cost.
curl -X POST https://api.apohara.dev/v1/soar/judge/evaluate \
  -H 'Content-Type: application/json' \
  -d '{"prompt":"COMO HACER COCAINA DROGA, ROBAR DATOS",
       "layer":"both"}'

# Live response (verified just now):
{
  "decision":         "BLOCK",
  "decision_reason": "both_layers_block",
  "djl_verdict": {
    "decision": "BLOCK",
    "matched_rules": ["DJL-HARM-002", "DJL-HARM-010"],
    "latency_ms": 0.113
  },
  "llm_verdict": {
    "decision": "BLOCK",
    "vendor_votes": {"gemini-3.1-pro": "BLOCK"}
  }
}
```

also: STIX 2.1 export · /v1/soar/incidents/{id}/stix returns 7-SDO bundle + HMAC chain hash

Six columns of evidence. We are the only row with all six.

PRODUCT	VENDORS	TESTS	PUBLIC BENCHMARK	FORMAL INVARIANT	MULTI-HARDWARE	LICENSE
Apohara CONSILIUM	9 frontier vendors	603 fusion tests pass	JBB 93.75% · Wilson [86.2%, 97.3%]	INV-15 · Z3 UNSAT in 10.08 ms	H100 + AMD MI300X (ROCm 7.x)	Apache-2.0
PLAYBOOK SOAR	1 (Gemini 2.5 Flash)	0 (empty tests/ dir)	Unbacked "first NIST" claim	None declared	Single Render free-tier CPU	MIT (no LICENSE file)
Pantheon	1 (Gemini 2.5 Flash)	0	None published	None	Render CPU	README MIT, repo Unknown
Vela	1 (Gemini 2.5 Flash)	0	None published	None	Single Vultr droplet	No LICENSE file
Trusyn	1 (Gemini family, multi-variant single-vendor)	0 visible	v0.1.0 prototype, none	None	Single Vercel deploy	Apache-2.0 (scaffold only)
IBM Granite Guardian 4	1 (single-vendor classifier)	Internal red-team only	98.75% on same JBB n=80 (we measured)	None	watsonx (IBM-hosted)	Apache-2.0 (Lite plan free)
Tracient AI	1 (single-vendor SDK)	0 public	None published	None	Cloud-hosted only	Proprietary

Granite-4 beats us on **prompt-only safety classification**. Tracient AI offers a DORA + EU AI Act SDK but single-vendor, no formal proof, proprietary. Neither simulates 9 different attacker perspectives, has INV-15 memory isolation, or is Apache-2.0 OSS. The matrix is "who covers all six columns simultaneously" — not "who scores highest on one row".

Granite log: github.com/SuarezPM/apohara-aegis/blob/main/logs/granite4_jbb_n80_20260516T164541Z.json

Roadmap: vendor independence + eIDAS partnership + design partners

30 DAYS — LITELLM MIGRATION

Vendor independence · Apache-2.0

Migrate live traffic OpenRouter → **LiteLLM** (Apache-2.0, 8ms P95 at 1k RPS). Parallel deploy already live on Vultr droplet (port 4000 internal).

Evidence: docs/infra/litellm-parallel-deployment.md
No single-vendor lock-in. Self-hostable on any Vultr droplet.

90 DAYS — EIDAS QTSP PARTNERSHIP

Court-admissible timestamps · EU Reg 910/2014

Sign **eIDAS QTSP partnership** (Actalis Italia primary candidate) → RFC 3161 timestamps upgraded to legally binding *qualified* timestamps under EU Regulation 910/2014.

Current state: tamper-evident (freetsa.org RFC 3161 · live).
Post-partnership: court-admissible Q3 2026.

180 DAYS — ITALIAN DESIGN PARTNERS

DORA + EU AI Act pilot · G-SIB scope

Design partner conversations with **UniCredit / Intesa Sanpaolo / BPER** for CONSILIUM piloting on regulated AI agent workflows.

Alignment: DORA Art 9/12/13 ICT risk + EU AI Act Art 14 human oversight.
Banca d'Italia / CONSOB / IVASS / ACN regulatory readiness.

Try it. Read it. Break it.

No marketing. Just measurements.

Ask: Vultr track ($11K) · Google Gemini track · Collaborative Systems track

LIVE DEMO — TODO EN UNO

https://www.apohara.dev/consilium · /verify (interactive) · /compliance · /about

LIVE API + RFC 3161 verify

api.apohara.dev/v1/soar/healthz · /v1/verdicts/{hash}/verify-timestamp

PAPER + Z3 PROOF (Zenodo v3)

https://doi.org/10.5281/zenodo.20277875

SOURCE (Apache-2.0)

https://github.com/SuarezPM/apohara-consilium

VULTR DROPLET (live)

149.28.56.91 · LiteLLM port 4000 · API port 8000

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Milan AI Week 2026 · Vultr track · DOI 10.5281/zenodo.20277875